

Introduction to Natural Language Processing

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Welcome

- Language as used by humans, naturally and effortlessly.
- This course is about building systems that work with human language.
- Systems that can read, write, translate and reason over text.
- And the models and data that make this possible.

Humans understand language
easily.

Why is it so hard for machines?

What Can NLP Systems Do?



Language Is Not Just Data

- Language is structured.
- Language is ambiguous.
- Language is contextual.
- Language varies across speakers and communities.

NLP systems learn from data, but the data reflects these linguistic properties.

Why Language Is Hard for Machines

Ambiguity exists everywhere.

- **Lexical ambiguity**

“I saw a bat.”

- **Structural ambiguity**

“I saw the man with the telescope.”

- **Pragmatic ambiguity**

“Can you pass the salt?”

Humans resolve these effortlessly. Machines need explicit models.

How Do NLP Systems Work?

- Represent language in a form machines can process.
- Learn patterns from data.
- Generalize to unseen examples.

Every NLP system makes assumptions about what matters in language.

Where Do These Assumptions Come From?

- From data.
- From task design.
- From insights about how language is structured and used.

Good NLP engineering depends on making the **right assumptions**.

Example: Machine Translation

Hindi: *raam ne kitaab padhee.*

Telugu: *raamuDu pustakam cadivaaDu.*

To translate correctly, a system must handle:

- word order
- case marking
- agreement

Why Multilingual NLP Is Hard

- Languages differ in structure.
- They differ in writing systems.
- They differ in how meaning is expressed.

Methods that work well for English may fail elsewhere.

Indian Languages in NLP

- Many Indian languages are morphologically rich.
- Word order can be flexible.
- Scripts encode sound and meaning differently.

These properties matter when designing NLP systems.

Code-Mixing Is Normal

- “Wait karo, main 5 minutes mein aa raha hoon.”
- Mixing languages and scripts is natural for humans.

For NLP systems, this raises difficult modeling questions.

Low-Resource Languages

- Many languages have limited digital data.
- Models trained on high-resource languages do not always transfer well.
- This creates unequal access to language technology.

What This Course Will Focus On

By the end of this course, you will:

- Understand core NLP problems and methods
- Develop a scientific and computational understanding of language
- Use this understanding to build NLP models
- Analyze the strengths and limitations of these models
- Be prepared for advanced NLP and ML topics

What This Course Will Not Do

- Blind use of APIs
- Treat language as just a string of characters
- Ignore linguistic and social context

Strong NLP systems start with correct assumptions.

Models change.

Data grows.

Understanding language remains essential.

Welcome to Introduction to NLP